
CHAPTER 1

SET THEORY

“BEFORE WE CAN DO STATISTICS WE HAVE TO DO PROBABILITY THEORY,
AND FOR THAT WE NEED SOME SET THEORY.”

1.1 Set definitions

Definition 1.1.1 (Set). A set is defined to be a collection of distinguishable objects or entities.

These objects can be human beings, cars, numbers, alphabets or any other elements. It is usually denoted by the upper case letters such as A, B, C and so on. Some symbols are reserved for specific sets, for example:

$\mathbb{N} = \{0, 1, 2, \dots\}$ = set of natural numbers;

$\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$ = set of all integers;

$\mathbb{R}$ = set of real numbers.

The symbol $x \in A$ is used to mean that ‘ x is an element of the set A ’ and $x \notin A$ is used to mean that ‘ x is not an element of the set A ’.

The set must be well defined in the sense that it must be possible at least in principle, to specify the set so that one can decide whether any given number does or does not belong to the set. The members of the set are distinct in the sense that repetition of

elements is not permitted in specifying the set. There are two ways in which a set A can be determined:

- (a) a list of objects that belong to A or
- (b) give a rule which indicates whether or not a given element belongs to A .

Example 1.1.1. If we say, ‘the numbers 2, 16, 9 belong to set A ’, then its written:

$$A = \{2, 16, 9\}.$$

Example 1.1.2. If we say, ‘the set B consists of real numbers between -2 and 2 ’, then its written: $B = \{x \mid -2 \leq x \leq 2\}$.

Exercise 1

- (a) Express in set notation: the set A of numbers which have square roots between 1 and 16.
- (b) Write out in full all the elements of the set $B = \{(x, y) \mid x \in \{1, 2, 3, 4\}, y = x^2\}$

Definition 1.1.2 (Universal set). An *universal set* contains all possible objects in consideration. It is usually denoted by letters U and S . The mathematicians knows it as an universal set, hence the notation U whereas in statistics jargon is referred to as the *sample space*, hence denoted by S .

Definition 1.1.3 (Empty set). An *empty set* is set that contains no elements. It is sometimes referred to as the *null set* and denoted by $\emptyset$.

Definition 1.1.4 (Subset). Given two sets A and B , A is said to be a *subset* of B if every element of A is also an element of B . It is written as $A \subset B$ and read as ‘ A is contained in B .’

Alternatively, it can be written as $B \supset A$ and read B contains A . If every element of A does not belong to B , then it is written as $A \not\subset B$ and read ‘ A is not a subset of B ’.

Definition 1.1.5 (Equal Sets). Two sets A and B are equal if and only if $A \subset B$ and $B \subset A$.

Definition 1.1.6 (Power set). Given a set A , the set of all subsets of A is called the *power set of A* . It is denoted by $\mathcal{P}(A)$.

Example 1.1.3. Let $A = \{1, 3, 5\}$, $B = \{1, 3, 5, 6, 9\}$ and $C = \{1, 2, 3, 4, 5\}$. Then it is clear that $A \subset B$, $B \not\subset C$ and $A \subset C$.

Example 1.1.4. Let $A = \{1, 2, 3\}$, then the power set of A is given by

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, A\}$$

Exercise 2

Let $A = \{x \mid 0 < x < 5\}$, $B = \{0, 5\}$, $C = \{1, 2, 3, 4\}$, $D = \{2, 4\}$ and $E = \{x \mid 1 \leq x \leq 4\}$. Which of the following statements are true and which are false?

(a) $C = E$

(b) $D \subset C$

(c) $D \in E$

(d) $E \not\subset A$

1.2 Set operations

The following set operations – intersection, union, complement and set difference are defined in this section.

1.2.1 Intersection

Definition 1.2.1 (Intersection). The *intersection* of any two sets, denoted by $A \cap B$, is the set that contains elements which belong to both sets. That is,

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

In cases where there are more than two sets considered, say n sets, their intersection is denoted by $A_1 \cap A_2 \cap \dots \cap A_n$ or simply $\bigcap_{i=1}^n A_i$.

Example 1.2.1. Consider the following sets of natural numbers: $P = \{x \mid 0 \leq x \leq 10\}$ and $Q = \{x \mid 5 < x < 20\}$. Find $P \cap Q$. Is $5 \in P \cap Q$?

Paying careful attention to the strict inequalities of the endpoints in set Q , it should be clear that

$$P \cap Q = \{x \mid 5 < x \leq 10\}.$$

And no, $5 \notin P \cap Q$.

Definition 1.2.2 (Disjoint sets). Two non-empty sets A and B such that $A \cap B = \emptyset$ are said to be *disjoint*. That is, disjoint sets have no elements in common.

1.2.2 Union

Definition 1.2.3 (Union). The *union* of two sets A and B , denoted by $A \cup B$, is the set which contains elements that belongs to A **or** B . That is,

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Note that the use of word **or** means in an inclusive sense – the union does not excludes the elements that belong to both A and B . From Example 1.2.1, the union includes all

elements from both set P and Q so that:

$$P \cup Q = \{x \mid 0 \leq x < 20\}.$$

In general one can have a union of more than two sets. For n sets $A_1, A_2, \dots, A_n$, the union of these is defined as, the set of all those elements which belong to at least one set out of these n sets. The union of n sets is denoted by $A_1 \cup A_2 \cup \dots \cup A_n$ or simply $\bigcup_{i=1}^n A_i$.

1.2.3 Complements

Definition 1.2.4 (Complement). Let S be the sample space (universal set in Math) and A a subset of U . The *complement* of A , denoted by $\bar{A}$ or A^c , is the set of all elements of S which are not in A . That is,

$$\bar{A} = \{x \mid x \in S \text{ and } x \notin A\}$$

1.2.4 Set difference

Definition 1.2.5 (Difference). For any two sets A and B , difference of A and B , denoted by $A - B$ is the set of all elements of A which do not belong to B . That is,

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}.$$

From this definition is easy to intuitively deduce that $\bar{A} = S - A$ and $A - B = A \cap \bar{B}$.

1.2.5 Properties of the set operations

Let S be the sample space and A & B the subsets of S . Then the following properties follows immediately from the definition of the different set operators.

$$(i) \quad A \cup B = B \cup A; \quad A \cap B = B \cap A.$$

- (ii) $A \cup A = A \cap A = A$.
- (iii) $(A \cup B) \cup C = A \cup (B \cup C)$.
- (iv) $(A \cap B) \cap C = A \cap (B \cap C)$.
- (v) $A \cup \emptyset = A$; $A \cap \emptyset = \emptyset$.
- (vi) $A \cup \bar{A} = S$; $A \cap \bar{A} = \emptyset$.
- (vii) $A \cap B \subset A$.
- (viii) $A \subset A \cup B$.

Exercise 3

Let S be the sample space which contains the alphabet letters. That is, $S = \{a, b, c, \dots, z\}$.

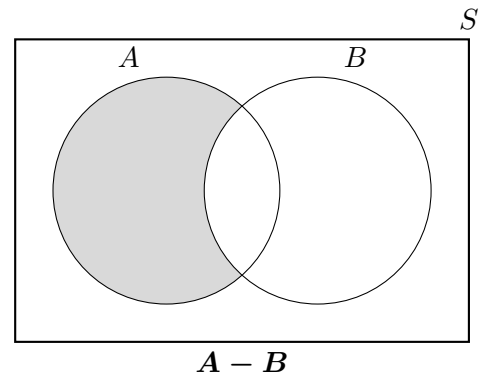
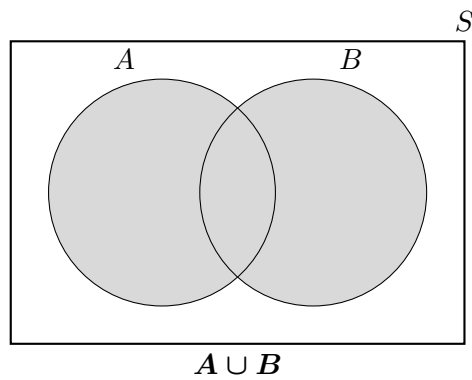
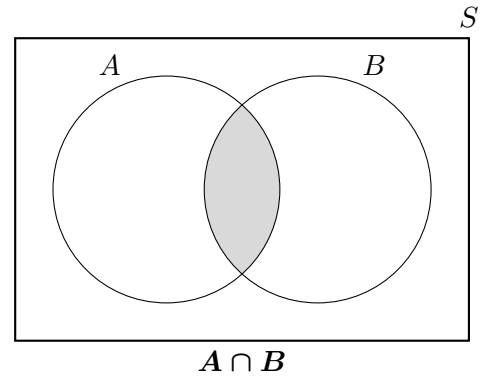
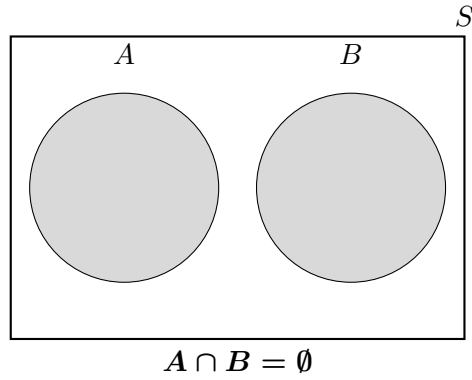
Let A be the set which contains vowels, B be the set of consonants and C contains the first 10 letters of the alphabet. Pick the TRUE and FALSE statements in the following list:

- (a) $A \cup B = S$
- (b) $A - C = \{o, u\}$
- (c) $\overline{(A \cup C)} = \bar{A} \cap \bar{C}$
- (d) $\bar{A} = B$.

1.3 Venn diagrams

Definition 1.3.1 (Venn diagram). A pictorial representation of sets is referred to as a *venn diagram*. The sample space is usually represented with a large rectangle and sets are denoted by various shapes within this rectangle.

Some of the set operators are illustrated below using venn diagrams.



Example 1.3.1. Illustrate with the use of Venn diagrams to show the following.

- (a) $A \cup B = A \cup (B \cap \bar{A})$.
- (b) $(\bar{A} \cap C) \cup (\bar{B} \cap A) = (A \cup C) \cap (\overline{B \cap A})$.

[Solution will be illustrated in class.]

Exercise 4

Let A, B and C be subsets of an universal set S . Draw Venn diagrams to determine which of the following are TRUE.

- (a) $A \cup B \cup C = S$.
- (b) $\overline{A \cup B} = \bar{A} \cap \bar{B}$.
- (c) $A \cap (B \cup \bar{C}) = (A \cap B) \cup (A \cap \bar{C})$.
- (d) $A \cap B \cap C = S \emptyset$

1.4 Algebra of Sets**1.4.1 Distributive laws**

Theorem 1.4.1.

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad (1.1)$$

Proof. Since the two sets are equal, we first need to show that

$A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C)$ and lastly that $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$.

(i) Let

$$x \in A \cap (B \cup C)$$

Then

$$x \in A \quad \text{and} \quad x \in (B \cup C).$$

This means that

$$x \in A \quad \text{and} \quad (x \in B \text{ or } x \in C).$$

That is,

$$x \in A \text{ and } x \in B \text{ or, } x \in A \text{ and } x \in C.$$

Hence

$$x \in (A \cap B) \cup (A \cap C).$$

- (ii) To prove the latter part, let $x \in (A \cap B) \cup (A \cap C)$. That is, $x \in (A \cap B)$ or $x \in (A \cap C)$. In words, x is in A and B or x is in A and C . Thus x is in A and x is in B or C . Hence $x \in A \cap (B \cup C)$.

Theorem 1.4.2.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \tag{1.2}$$

Proof. Left as an exercise for the students.

1.4.2 De Morgan's laws

Theorem 1.4.3.

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \tag{1.3}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B} \tag{1.4}$$

Example 1.4.1. Show that $A \subset B$ if and only if $A = A \cap B$.

Suppose that $A \subset B$. We know that $A = A \cap A$ and therefore

$$A \cap A \subset A \cap B$$

But we also know that $A \cap B \subset A$. Hence

$$A = A \cap B.$$

Conversely we have

$$A = A \cap B \subset B.$$

Example 1.4.2. Show that $(A - B) - C = A - (B \cup C)$.

$$\begin{aligned} (A - B) - C &= (A - B) \cap \bar{C} \\ &= (A \cap \bar{B}) \cap \bar{C} \\ &= A \cap (\overline{B \cup C}) \\ &= A - (B \cup C). \end{aligned}$$

Example 1.4.3. Show that $A - (B \cup C) = (A - B) \cap (A - C)$.

$$\begin{aligned} A - (B \cup C) &= A \cap (\overline{B \cup C}) \\ &= A \cap (\bar{B} \cap \bar{C}) \\ &= (A \cap \bar{B}) \cap (A \cap \bar{C}) \\ &= (A - B) \cap (A - C). \end{aligned}$$